Narek Harutyunyan

Website | Google Scholar | narek_harutyunyan@brown.edu | +1 (857) 404-2113

Education

University of California, Berkeley **Starting Fall 2026**
Incoming Ph.D. Student in EECS Berkeley, CA, USA

Brown University **September 2022 – May 2026**
Bachelor of Science in Computer Engineering | GPA: 4.0/4.0 Providence, RI, USA

Relevant Courses: *Learning and Sequential Decision Making (PhD level) | Deep Learning | Computer Systems | Artificial Intelligence | Coordinated Mobile Robotics (PhD level) | Digital Electronics Systems Design | Design of Computing Systems | Linear System Analysis | Communication Systems*

Skills

Programming Languages: Python, C, C++, Java, JavaScript
Frameworks: ROS, Gazebo, IsaacSim & IsaacLab, PyBullet, Mujoco, TensorFlow, PyTorch, NumPy
Developer Tools: Git, Linux, RViz,
Robotics Toolchain: Fusion 360, MATLAB, Vicon, OptiTrack, Crazyflie, Unitree Go1, Boston Dynamics Spot, Booster T1 Humanoid, iRobot Create3 (Roomba), TurtleBot, custom-built quadrotors

Research Experience

Undergraduate Teaching and Research Awards (UTRA) **May 2023 – Present**
Brown University | Prof. Nora Ayanian | ACT Lab *Providence, RI, USA*

- Design reinforcement learning policies in IsaacLab for cooperative object transport with swarms of Crazyflie drones, enabling robust multi-agent coordination.
- Characterize quadcopter downwash interactions using Particle Image Velocimetry and MATLAB, improving close-proximity flight stability and formation safety.
- Engineer an aerial painting quadrotor with custom components and Python wrappers, enabling hands-on experimentation for 75+ students in a university robotics curriculum.
- Built a real-time LED + audio-responsive quadrotor choreography system in Python/C, enabling synchronized drone performances that react dynamically to music.

Summer Undergraduate Research Fellowships (SURF) **May – September 2025**
California Institute of Technology | Prof. Soon-Jo Chung | ARCL Lab *Pasadena, CA, USA*

- Engineered a high-fidelity IsaacLab simulation of the Unitree Go1 quadruped on a moving ground plane, emulating ship-like dynamics for robust policy training.
- Developed a certified RL algorithm with contraction theory for stable quadruped locomotion and handstands on a moving platform, achieving 100% deployment success.
- Built a custom SDK for real-robot deployment and validated Unitree Go1 policies under disturbances, including wind on a motion platform and tests on an inflatable boat.
- Designed and deployed reinforcement learning policies for locomotion and fault recovery on the humanoid robot Booster T1, demonstrating reliable performance on hardware.

Robotics Institute Summer Scholar Program (RISS) **May 2024 – July 2025**
Carnegie Mellon University | Prof. Sebastian Scherer | AirLab *Pittsburgh, PA, USA*

- Refactored the MapEx framework (probabilistic frontier-based exploration with predictive maps) into Gym, added an IoU evaluation metric, and fixed algorithmic issues like agent backtracking.
- Proposed and led MapExRL, a human-inspired RL exploration framework using frontier-based planning, global map predictions, and uncertainty modeling for long-horizon reasoning.
- Achieved up to 18.8% performance gains over state-of-the-art exploration baselines on real-world indoor maps with MapExRL.

Teaching Experience

Creating Art with Teams of Robots | *Head TA* **Spring 2024, Fall 2024, Spring 2026**

- Guided 75+ students weekly in labs and final projects, ensuring successful execution through hands-on mentorship and teamwork.
- Developed and evaluated final project concepts using Crazyflie quadcopters, quadrupeds, and ground robots, incorporating the innovative painter drone designed in previous ACT Lab research.

Deep Learning | *Teaching Assistant* **Fall 2025**

- Support a 120+ student course by designing labs and assignments on deep learning, and leading biweekly workshops on emerging AI trends.
- Hold weekly office hours and mentor final project teams, providing personalized guidance on technical challenges and the design of deep learning solutions in robotics and AI.

Introduction to Engineering | *Teaching Assistant* **Fall 2023**

- Coached 40+ students in engineering projects, including chair construction, electrical door locking mechanisms, and electric guitar building.
- Led workshops for 100+ students on MATLAB, CAD (Fusion 360), prototyping (3D printing, laser cutting), machine shop tools, and Arduino-based electronics.

Publications and Posters

[1] V. Zinage*, **N. Harutyunyan***, E. Verheyden*, F. Hadaegh, S.-J. Chung, "**ContractionPPO**: Certified Reinforcement Learning via Differentiable Contraction Layers," in IEEE Robotics and Automation Letters (RA-L), under review | |

[2] **N. Harutyunyan***, B. Moon*, S. Kim, C. Ho, A. Hung, S. Scherer, "**MapExRL**: Human-Inspired Indoor Exploration with Predicted Environment Context and Reinforcement Learning," presented at ICRA 2025 Workshop on Structured Learning for Efficient, Reliable, and Transparent Robots, to appear at the IEEE International Conference on Advanced Robotics (ICAR), 2025 | |

[3] C. Ho*, S. Kim*, B. Moon, A. Parandekar, **N. Harutyunyan**, C. Wang, K. Sycara, G. Best, S. Scherer, "**MapEx**: Indoor Structure Exploration with Probabilistic Information Gain from Global Map Predictions," in IEEE International Conference on Robotics and Automation (ICRA), 2025 | |

[4] A. Kiran, **N. Harutyunyan**, N. Ayanian, K. Breuer, "**Downwash Dynamics**: Impact of Quadrotor Separation on Forces, Moments, and Velocities for Dense Formation Flight," in AIAA Aviation Forum and ASCEND, 2024 |

[5] O. Heng*, **N. Harutyunyan***, J. Dhanda*, "RoboChaser," presented at Brown University Deep Learning Day — TAs' Choice Award for Best Project among 100+ projects, 2024 |

[6] A. Min*, R. Hossain*, H. Izhar*, **N. Harutyunyan***, "Advancements in Multi-Robot Systems," presented at Yale Northeast Robotics Colloquium (NERC), 2023 |

** denotes equal contribution*

Awards and Honors

Tau Beta Pi Engineering Honor Society **2024 – Present**
Inducted member recognizing academic excellence in engineering *Providence, RI, USA*

Summer Undergraduate Research Fellowships (SURF) **2025**
California Institute of Technology *Pasadena, CA, USA*

Robotics Institute Summer Scholar (RISS) **2024**
Carnegie Mellon University *Pittsburgh, PA, USA*

Undergraduate Teaching and Research Award (UTRA) **2023, 2024**
Brown University *Providence, RI, USA*

Huys Scholar **Awarded 2023, 2024**
Huys Foundation STEM Scholarship *(5% selection rate)* *West Hollywood, CA, USA*